

Statistics Exam Master Guide

Multiple Linear Regression & Time Series

How to read questions · Which method to use · Why it works Common pitfalls ·
Different phrasings · Worked decision logic

This guide walks you through every type of question that appears in exams on Multiple Linear Regression (MLR) and Time Series. For each topic you will find: what the question is really asking, which method to choose and why, step-by-step decision logic, the correct interpretation, pitfalls that cost marks, and different phrasings that mean the same thing.

PART 1 — MULTIPLE LINEAR REGRESSION (MLR)

1.1 How to Read an MLR Question

Before touching SPSS, read the question twice and identify:

What to identify	How it appears	What you do
Dependent variable Y	"Regress X on Y" / "explain Y" / "Y as a function of"	Put in Dependent box
Continuous predictors	Age, income, price — no category labels	Enter directly
Categorical predictor	"Region", "type", "group" with labels like North/South	Create dummy variables (k–1)
Log transformation	"ln(X)", "log(X)" in variable name	Variable already logged, or use Transform → Compute
Interaction term	"effect depends on Z", "moderating", "X×Z"	Create product variable X×Z

1.2 Dummy Variables

A dummy (indicator) variable is 0/1 and converts a categorical variable with k categories into k–1 binary variables. The omitted category is the reference group — all coefficient interpretations are made relative to it.

Rule: k categories → create k–1 dummies

Categories	Dummies to create	Reference (omitted)
North, South, Center, East (4 groups)	D_North, D_South, D_East (3 dummies)	Center
Male, Female (2 groups)	D_Female (1 dummy)	Male
Low, Medium, High (3 groups)	D_Medium, D_High (2 dummies)	Low

In SPSS: Transform → Compute Variable

`D_North = (Region = 1) [if North is coded 1 in the data]`

`D_South = (Region = 2)`

`D_East = (Region = 4) [skip 3 = Center → it is the reference]`

Theoretical model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 \cdot D_North + \beta_3 \cdot D_South + \beta_4 \cdot D_East + \varepsilon$$

Interpretation of β_2 : On average, North regions have β_2 units MORE Y

than Center regions, holding all else constant.

⚠ Common Pitfalls

- Never create k dummies — always $k-1$. Including all k causes perfect multicollinearity (dummy trap). Always omit one reference category.
- "Baseline group", "comparison group", "omitted category", "reference category" — all mean the same thing.
- The question may say "use Center as reference" — that means do NOT create a dummy for Center.

1.3 Log Transformations and Interpretation

Model type	Equation	Interpretation of β
Linear (standard)	$Y = \alpha + \beta X + \varepsilon$	1-unit rise in $X \rightarrow \beta$ unit change in Y
Linear-log	$Y = \alpha + \beta \cdot \ln(X) + \varepsilon$	1% rise in $X \rightarrow \beta/100$ unit change in Y
Log-linear	$\ln(Y) = \alpha + \beta X + \varepsilon$	1-unit rise in $X \rightarrow \beta \times 100$ % change in Y
Log-log (elasticity)	$\ln(Y) = \alpha + \beta \cdot \ln(X) + \varepsilon$	1% rise in $X \rightarrow \beta$ % change in Y

Linear-Log: Most Tested Type

- Most common exam type: Linear-log. $Y = \alpha + \beta \cdot \ln(X) + \varepsilon$
- Example: $\beta = -16.04$, X is $\ln(\text{NO}_x)$.
- Answer: 1% increase in $\text{NO}_x \rightarrow -16.04/100 = -0.16$ unit change in Y .
- If Y is measured in thousands of dollars: $-\$160$ per 1% increase in NO_x .
- Watch for ' $\ln(X)$ ' or ' $\log(X)$ ' in the variable name in SPSS output.

1.4 Interpreting Coefficients — Decision Table

Is X logged?	Is Y logged?	Interpretation of β
No	No	1 extra unit of $X \rightarrow \beta$ extra units of Y (ceteris paribus)
Yes — $\ln(X)$	No	1% extra in $X \rightarrow \beta/100$ extra units of Y
No	Yes — $\ln(Y)$	1 extra unit of $X \rightarrow \beta \times 100$ % change in Y
Yes — $\ln(X)$	Yes — $\ln(Y)$	1% extra in $X \rightarrow \beta$ % change in Y (elasticity)
Dummy (0/1)	No	Being in group D $\rightarrow \beta$ extra units of Y vs. reference group

1.5 Testing Individual Coefficient Significance

$H_0: \beta = 0$ (variable has no effect on Y)

$H_1: \beta \neq 0$ (variable does affect Y)

Decision: p-value (Sig. column) < 0.05 → Reject H_0 → significant at 5%

p-value ≥ 0.05 → Fail to reject H_0 → not significant

⚠ Common Pitfalls

- "Significant" means statistically distinguishable from zero — not necessarily large or important.
- "Is X relevant?", "Does X matter?", "Is β significantly different from zero?", "Test the effect of X" — all ask for the t-test and p-value.

1.6 Joint F-test — Testing a Group of Variables Together

Use the joint F-test when the question asks whether a GROUP of variables is significant together, not just one at a time.

Trigger phrases — use joint F-test when you see these

- "Are the region dummies jointly significant?"
- "Do the seasonal dummies together improve the model?"
- "Test whether D_1, D_2, D_3 together add explanatory power."
- "Is the set of interaction terms significant?"
- Any question testing MORE THAN ONE variable simultaneously.

Step	What you do	Why
1	Run UNRESTRICTED model — all variables included	Gives SSE_u and R^2_u
2	Run RESTRICTED model — remove the variables being tested	Gives SSE_r and R^2_r
3	J = number of variables removed	J = numerator degrees of freedom
4	Read n (observations) and k (predictors in unrestricted model)	For denominator df
5	Compute F and compare to p-value	Decision

Option A (using SSE from ANOVA tables):

$$F = [(SSE_r - SSE_u) / J] \div [SSE_u / (n - k - 1)]$$

Option B (using R^2 from Model Summary):

$$F = [(R^2_u - R^2_r) / J] \div [(1 - R^2_u) / (n - k - 1)]$$

Decision: Use the p-value (Sig.) from SPSS ANOVA table if available.

Or compare your computed F to the critical value from the F-table.

⚠ Common Pitfalls

- J = number of RESTRICTIONS (variables removed), not total variables in model.
- Restricted model has FEWER variables — its SSE is always LARGER.
- $n-k-1$ uses k from the UNRESTRICTED model.
- "Test if dummies are jointly significant" = run restricted model WITHOUT those dummies.

1.7 Residual Plot Analysis

Problem	What you see in the plot	Assumption violated
Heteroskedasticity	Spread of residuals grows or shrinks — fan shape (wider on one side)	A2: $\text{Var}(\epsilon) = \sigma^2$ (constant variance)
Non-linearity	Residuals curve upward or downward — systematic arc pattern	A1: model correctly specified (linearity)
Outliers	One or a few points far above/below the rest	Distort estimates — identify and report

Good vs. Bad Residual Plot

- Good residual plot: random cloud of points centred near zero. No pattern, no fan, no curve.
- Bad residual plot: any systematic shape — V-shape, cone/fan, curve, or large isolated points.
- You do NOT need to quantify — describe what you SEE and name the assumption violated.

⚠ Common Pitfalls

- Do not say 'residuals are not normal' from a residual-vs-fitted plot alone. Normality requires a histogram or Q-Q plot.
- Heteroskedasticity → standard errors are wrong → t-tests and p-values are unreliable. This is why we follow up with the White test.

1.8 The White Test for Heteroskedasticity

H_0 : homoskedasticity — $\text{Var}(\epsilon) = \sigma^2$ (constant)

H_1 : heteroskedasticity — variance is NOT constant

Step 1: Run original model. Save residuals $\hat{e}$ (Save → Unstandardized Residuals).

Step 2: Create $\hat{e}^2$ via Transform → Compute Variable: $\text{RES_SQ} = \text{RES_1} ** 2$

Step 3: Run auxiliary regression:

$\hat{e}^2 = \alpha_0 + (\text{all original } X\text{'s}) + (X^2 \text{ for each continuous } X)$

+ $(X_i \times X_j \text{ for each pair of continuous } X\text{'s}) + u$

Do NOT square dummies. Do NOT include dummy $\times$ dummy terms.

Step 4: $W = n \times R^2_{\text{aux}} \sim \chi^2(df)$

df = number of regressors in auxiliary regression (excluding constant)

Step 5: If $W > \text{critical value}$ OR $p < 0.05 \rightarrow \text{Reject } H_0 \rightarrow \text{heteroskedasticity}$

⚠ Common Pitfalls

- The df for the White test is NOT 1. Count all terms in the auxiliary regression: original X's + squared continuous X's + cross-products.
- Example with 2 continuous X's: $X_1, X_2, X_1^2, X_2^2, X_1 \times X_2 \rightarrow df = 5$.
- Dummies: $D^2 = D$ for a 0/1 variable, so squaring adds nothing. Exclude them.
- Always state df in your answer: $W \sim \chi^2(5)$. Different questions have different df.

1.9 Omitted Variable Bias

If a relevant variable Z is left out of the model, the estimated coefficient on the included variable X will be biased. The direction of the bias:

$\text{sign}(\text{bias on } \beta_X) = \text{sign}(\beta_Z) \times \text{sign}(r(X, Z))$

β_Z = true effect of omitted variable Z on Y

$r(X, Z)$ = correlation between included X and omitted Z

Bias $> 0 \rightarrow$ estimate is TOO HIGH (upward bias)

Bias $< 0 \rightarrow$ estimate is TOO LOW (downward bias)

β_Z (effect of Z on Y)	$r(X, Z)$ (correlation with X)	Bias on β_X	Estimate is...
Positive (+)	Positive (+)	Positive $\rightarrow$ upward bias	Too high
Positive (+)	Negative (–)	Negative $\rightarrow$ downward bias	Too low
Negative (–)	Positive (+)	Negative $\rightarrow$ downward bias	Too low
Negative (–)	Negative (–)	Positive $\rightarrow$ upward bias	Too high

Worked Example

- Worked example: estimating effect of $\ln(\text{NOx})$ on housing values, omitting Indus (industry).
- $\beta_{\text{Indus}} < 0$ (more industry $\rightarrow$ lower house values).
- $r(\ln(\text{NOx}), \text{Indus}) > 0$ (areas with high NOx also tend to have more industry).
- Bias = $(-) \times (+) = \text{negative} \rightarrow$ estimate of $\beta_{\ln\text{NOx}}$ is TOO LOW (overestimates the negative effect).

⚠ Common Pitfalls

- "Biased downward" means the estimate is too small (too negative for a negative coefficient).
- If the correlation is not given directly, reason from context: do areas with high NOx tend to also have more industry? If yes, correlation is positive.
- "Omitted variable bias" and "specification error" mean the same thing.

1.10 VIF and Multicollinearity

$$\text{VIF}_j = 1 / (1 - R^2_j)$$

$$\text{Tolerance}_j = 1 - R^2_j = 1 / \text{VIF}_j$$

$$R^2_j = R^2 \text{ from regressing } X_j \text{ on all other } X\text{'s (auxiliary regression)}$$

Rule of thumb: $\text{VIF} > 5 \rightarrow$ concerning multicollinearity

$\text{VIF} > 10 \rightarrow$ severe multicollinearity

$\text{Tolerance} < 0.20 \rightarrow$ concerning

VIF	Tolerance	Interpretation
1.0	1.00	No multicollinearity — X is uncorrelated with all other X's
1 to 5	0.20 to 1.00	Acceptable — moderate correlation, no real problem
5 to 10	0.10 to 0.20	Concerning — check estimates carefully
> 10	< 0.10	Severe — coefficient estimates are unreliable

- In SPSS: Analyze $\rightarrow$ Regression $\rightarrow$ Linear $\rightarrow$ Statistics $\rightarrow$ tick Collinearity diagnostics.
- VIF and Tolerance appear in the Coefficients table.
- If asked to compute VIF manually: run auxiliary regression of X_j on all other X's, read R^2 , compute $\text{VIF} = 1/(1 - R^2)$.

1.11 Quick Reference — Question Phrasing → Method

If the question says...	It means...	Method
"Interpret the coefficient of $\ln(X)$ "	Effect of 1% change in X	$\beta/100$ units change in Y
"Test if Region is significant"	All region dummies jointly	Joint F-test
"Is β_2 significantly different from zero?"	Individual significance	p-value from Coefficients table
"Test for heteroskedasticity"	Unequal error variance	White test: $W = n \times R^2_{\text{aux}}$
"What is the bias if Z is omitted?"	Direction of bias	$\text{sign}(\beta_Z) \times \text{sign}(r(X, Z))$
"What is the reference category?"	Omitted dummy group	State which group has no dummy
"VIF = 2.5 — what does this mean?"	Multicollinearity severity	Compare to threshold of 5
"Write the theoretical model"	Full symbolic equation	$Y = \beta_0 + \beta_1 X_1 + \beta_2 D_1 + \dots$

PART 2 — TIME SERIES REGRESSION

2.1 The Three-Step Framework — Always in This Order

The Three-Step Framework — never skip or reorder these

- STEP 1 — STATIONARITY: Run the Dickey-Fuller test on every variable. Non-stationary series must be differenced before use.
- STEP 2 — FIT THE MODEL: Choose static, DL(q), or AR(1). For DL(q) use top-down lag selection.
- STEP 3 — CHECK AUTOCORRELATION: Run the LMSC test on the model residuals.

⚠ Common Pitfalls

- You cannot skip Step 1. Using non-stationary series produces spurious results: high R^2 and significant t-values that have no real meaning.
- Steps must go in order: stationarity → fit → check. Never fit first.

2.2 Step 1 — Stationarity: The Dickey-Fuller Test

A stationary series has a constant mean and variance. A non-stationary series drifts — its mean changes over time. The Dickey-Fuller (DF) test formally tests whether a unit root is present.

Choose DF Version 1 or Version 2:

Version	Regression run	When to use
Version 1 — no trend	$\Delta Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \varepsilon_t$	Time plot shows NO visible upward or downward trend
Version 2 — with trend	$\Delta Y_t = \alpha_0 + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t$	Time plot shows a clear trend OR when unsure (safer default)

$H_0: \alpha_1 = 0$ (unit root — series is NON-STATIONARY)

$H_1: \alpha_1 < 0$ (series IS stationary) ← one-sided test

Special critical values at 5% (NOT the standard t-table):

Version 1 (no trend): -2.86

Version 2 (with trend): -3.41

Decision: t on Y_{t-1} more negative than critical value → Reject H_0 → STATIONARY

t on Y_{t-1} less negative (closer to 0) → Fail to reject → UNIT ROOT

Example A: t = -1.64, critical = -3.41

-1.64 > -3.41 → fail to reject H_0 → NON-STATIONARY → take first differences

Example B: $t = -4.20$, critical = -3.41

$-4.20 < -3.41 \rightarrow$ reject $H_0 \rightarrow$ STATIONARY $\rightarrow$ use in levels

DF result	Action	Next step
Stationary (reject H_0)	Use series in levels	Proceed to fit model
Non-stationary (fail to reject)	Take $\Delta Y = Y_t - Y_{t-1}$	Run DF again on ΔY
ΔY is stationary (reject H_0)	Series is $I(1)$ — use first differences	Proceed to fit model
ΔY still non-stationary	Rare — take second differences	Series may be $I(2)$

⚠ Common Pitfalls

- NEVER use the standard t-table (± 1.96) for the DF test. The DF distribution is non-standard. Always use -2.86 (V1) or -3.41 (V2) at 5%.
- The p-value SPSS shows for the LAG variable uses the wrong distribution — ignore it. Compare the t-statistic directly to the special critical value.
- Rejecting H_0 in the DF test means the series IS stationary — opposite of most tests.
- The DF test is one-sided: you only reject when the t-statistic is sufficiently NEGATIVE.

2.3 Step 2 — Which Model to Fit?

Model	Equation	When to use
Static	$Y_t = \alpha + \beta X_t + \varepsilon_t$	Question specifies it, or top-down DL gives $q = 0$
DL(q) — Distributed Lag	$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \dots + \beta_q X_{t-q} + \varepsilon_t$	Question asks for DL; effects of X are delayed
AR(1) — Autoregressive	$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$	Question specifies AR(1); autocorrelation found in DL residuals

2.4 DL(q) — Top-Down Lag Selection

Setup in SPSS — create lag variables via Transform $\rightarrow$ Compute Variable:

$LAG1_X = \text{LAG}(X, 1)$ $LAG2_X = \text{LAG}(X, 2)$ $LAG3_X = \text{LAG}(X, 3)$

Top-down procedure:

1. Fit $DL(q_{\max})$. Look at p-value of the HIGHEST lag ($LAGq_X$).
2. If $p > 0.05$: not significant $\rightarrow$ drop it $\rightarrow$ fit $DL(q-1)$.

3. If $p \leq 0.05$: significant $\rightarrow$ STOP. Current q is optimal.

4. Repeat until highest remaining lag is significant.

IMPORTANT: Never drop an intermediate lag.

If lag 2 is significant but lag 1 is not $\rightarrow$ keep BOTH.

You cannot have lag 2 without lag 1 in the model.

Total multiplier = $\beta_0 + \beta_1 + \beta_2 + \dots + \beta_q$

= long-run effect of a permanent 1-unit increase in X

Immediate effect (impact multiplier) = β_0 only

⚠ Common Pitfalls

- $q_{\max}$ is usually given in the question. If not, 3 or 4 is a common default.
- Each lag costs one observation. DL(3) with $n = 100$ gives only 97 usable rows.
- "Optimal lag length", "best DL model", "chosen q " — all mean the same thing.
- Do not confuse total multiplier (long-run, sum of all β) with β_0 (immediate effect only).

2.5 AR(1) — Autoregressive Model

Model: $Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$

β = immediate effect: 1-unit rise in X_t raises Y_t by β , holding Y_{t-1} fixed

γ = persistence: how much of last period's Y carries into this period

Requires $|\gamma| < 1$ for the model to be valid

Total multiplier = $\beta / (1 - \gamma)$

= long-run effect of a permanent 1-unit increase in X

Example: $\beta = -0.031$, $\gamma = 0.52$

Total multiplier = $-0.031 / (1 - 0.52) = -0.031 / 0.48 = -0.065$

⚠ Common Pitfalls

- AR(1) with remaining autocorrelation $\rightarrow$ coefficient estimates are BIASED. DL with autocorrelation $\rightarrow$ only INEFFICIENT (standard errors wrong, but β 's still correct). This is why always run LMSC after fitting AR(1).
- γ must be strictly between -1 and $+1$. If $|\gamma| \geq 1$ the series is non-stationary.
- "Persistence coefficient", "coefficient on lagged Y ", "AR coefficient" — all refer to γ .

2.6 Step 3 — LMSC Autocorrelation Test

$H_0: \rho = 0$ (no autocorrelation – A3 satisfied)

$H_1: \rho \neq 0$ (autocorrelation present – A3 violated)

Step 1: Fit original model. Save residuals: Save → Unstandardized Residuals → RES_1

Step 2: Create lagged residual: Transform → Compute → LAG_RES1 = LAG(RES_1, 1)

Step 3: Run auxiliary regression:

Dependent = RES_1

Independent = ALL original X's + LAG_RES1

Step 4: From Model Summary: read R^2_{aux}

From ANOVA table: $n_{aux} = \text{Total df} + 1$

Step 5: $LM = n_{aux} \times R^2_{aux} \sim \chi^2(1)$ under H_0

Critical value at 5%: 3.841

If $LM > 3.841$ (or $p < 0.05$) → Reject H_0 → autocorrelation present

If $LM \leq 3.841$ (or $p \geq 0.05$) → Fail to reject → A3 plausibly satisfied

⚠ Common Pitfalls

- The auxiliary regression MUST include ALL original X predictors, not just the lagged residual.
- For DL(q): auxiliary regression includes all original X's AND all lagged X's from the DL model, plus LAG_RES1.
- "Test for autocorrelation", "check A3", "Breusch-Godfrey test", "LMSC test" — all the same.
- Durbin-Watson (DW) is INVALID when Y_{t-1} appears in the model. Always use LMSC.

2.7 Critical Pitfall Grid — Rejecting H_0 Means Opposite Things!

Test	H_0 means...	Reject H_0 means...	Fail to reject means...
Dickey-Fuller	Series HAS unit root (NON-STATIONARY)	Series IS stationary ✓	Non-stationary — difference it
LMSC	NO autocorrelation (A3 satisfied)	Autocorrelation IS present — fix model	A3 plausibly satisfied ✓
Individual t-test	$\beta = 0$ (variable has NO effect)	Variable DOES have a significant effect	Cannot conclude variable matters
Joint F-test	All tested $\beta = 0$ (group has NO effect)	Group jointly significant	Cannot conclude group matters
White test	Homoskedasticity (A2 satisfied)	Heteroskedasticity present — A2 violated	A2 plausibly satisfied ✓

2.8 Deterministic vs. Stochastic Trends — Different Fix!

Feature	Deterministic trend	Stochastic trend (unit root)
Mathematical source	λt term — mean grows linearly	$\rho = 1$ — each shock is permanent
Pattern in time plot	Smooth, predictable upward drift	Erratic, wanders unpredictably
Variance over time	Constant	Grows without bound
After a shock	Series returns to the trend line	Series NEVER recovers
Correct fix	Add t (time) as regressor	Take first differences ΔY
Wrong fix → consequence	First-differencing creates artificial negative autocorrelation	Detrending does not remove a random walk

Decision rule: deterministic vs. stochastic

- How to distinguish: look at the DF test result.
- DF fails to reject H_0 (unit root confirmed) → stochastic trend → take differences.
- DF rejects H_0 (stationary) but time plot still shows a trend → deterministic trend → add t .

2.9 Time Series — Question Phrasings Reference

If the question says...	It means...	Method
"Is the series stationary?"	Does it have a unit root?	Dickey-Fuller test
"Test for a unit root"	Is $\rho = 1$?	Dickey-Fuller test
"Levels or differences?"	What is the stationarity property?	Run DF, then decide
"Determine optimal lag length"	What is the best q for $DL(q)$?	Top-down procedure
"Long-run effect of X on Y "	Total multiplier	$DL: \sum \beta_k \mid AR: \beta/(1-\gamma)$
"Immediate effect of X on Y "	Impact multiplier (current period only)	β_0 — coefficient on X_t only
"Test for autocorrelation"	Is $A3$ violated?	LMSC test: $LM = n \times R^2_{aux}$
"Is the model well-specified?"	Check all assumptions including $A3$	LMSC + residual plot
"Series is $I(1)$ "	Non-stationary in levels, stationary in differences	Use first differences
"Spurious regression risk"	Non-stationary series used in levels	Run DF first, then decide
"Disturbance term carries memory"	Autocorrelation is present	LMSC test

2.10 SPSS Step-by-Step Workflow

Step	SPSS action	Record
1a. Plot	Graphs → Chart Builder → Line → drag variable and TIME to axes	Does it trend? Wander?
1b. DF on levels	Compute $\text{DIFF_Y} = Y - \text{LAG}(Y,1)$ and $\text{LAG_Y} = \text{LAG}(Y,1)$. Regression: DV = DIFF_Y, IV = LAG_Y [+ TIME for Version 2]	t on LAG_Y. Compare to $-2.86/-3.41$
1c. DF on differences (if unit root found)	Compute $\text{DIFF2_Y} = \text{DIFF_Y} - \text{LAG}(\text{DIFF_Y},1)$ and $\text{LAG_DIFFY} = \text{LAG}(\text{DIFF_Y},1)$. Regression: DV = DIFF2_Y, IV = LAG_DIFFY [+ TIME for V2]	t on LAG_DIFFY. Compare to $-2.86/-3.41$
2a. Lag variables	Transform → Compute: $\text{LAG1_X} = \text{LAG}(X,1)$, $\text{LAG2_X} = \text{LAG}(X,2)$, etc.	Note observations lost
2b. Fit model	Regression with chosen variables (static / DL / AR)	All β , R^2 , n
3a. Save residuals	Regression dialog → Save → Unstandardized Residuals	Variable RES_1 created
3b. Lag residual	Transform → Compute: $\text{LAG_RES1} = \text{LAG}(\text{RES_1}, 1)$	Variable LAG_RES1 created
3c. Auxiliary regression	Regression: DV = RES_1, IV = all original X's + LAG_RES1	R^2 and n from Model Summary + ANOVA
3d. LMSC	$\text{LM} = n_{\text{aux}} \times R^2_{\text{aux}}$. $n_{\text{aux}} = \text{Total df} + 1$ from ANOVA	Compare to 3.841. State conclusion.

PART 3 — MASTER PITFALL & TRICK PHRASING LIST

3.1 The Top 15 Mistakes — and How to Avoid Them

1. Using p-value for DF test	SPSS's Sig. for the LAG variable uses the wrong distribution. Always compare the t-statistic directly to -2.86 (V1) or -3.41 (V2). Never use the p-value.
2. Reject H_0 in DF = non-stationary	It is the OPPOSITE: rejecting H_0 in the DF test means the series IS stationary. H_0 is the unit root (non-stationarity).
3. Creating k dummies instead of k-1	For k categories, create exactly k-1 dummies. Including all k causes the dummy trap (perfect multicollinearity). Always omit one reference category.
4. Using standard t-table for DF	The DF t-statistic does not follow a standard t-distribution. Values like ± 1.96 do not apply. Use -2.86 or -3.41 only.
5. Missing X's in LMSC auxiliary regression	The LMSC auxiliary regression regresses residuals on ALL original X's PLUS the lagged residual. Missing any original X gives the wrong R^2 and wrong LM.
6. Total multiplier = β_0 only	DL(q): total multiplier = $\beta_0 + \beta_1 + \dots + \beta_q$ (ALL lag coefficients). AR(1): total multiplier = $\beta/(1-\gamma)$. β_0 alone is only the immediate effect.
7. Using DW for AR(1) models	Durbin-Watson is INVALID when Y_{t-1} appears as a regressor. Always use the LMSC test.
8. Differencing when trend is deterministic	Only difference when DF confirms a unit root ($\rho = 1$). If the trend is deterministic ($ \rho < 1$, $\lambda \neq 0$), add t as a regressor instead. Differencing a deterministic trend creates artificial negative autocorrelation.
9. Dropping intermediate insignificant lags	In top-down selection, only drop the HIGHEST lag while insignificant. Never drop lag 1 while keeping lag 2 — the lag structure must be contiguous.
10. Squaring dummies in the White test	$D^2 = D$ for a 0/1 variable, so squaring dummies adds nothing. Only square continuous variables. Exclude dummy $\times$ dummy cross-products too.
11. Confusing significant with large	A coefficient can be statistically significant ($p < 0.05$) but economically negligible. Always comment on both significance AND the size/direction of the effect.

12. Wrong sign for omitted variable bias	Bias = $\text{sign}(\beta_Z) \times \text{sign}(r(X, Z))$. Two negatives give positive bias (upward). Draw a sign table if in doubt.
13. Ignoring reference category in interpretation	The coefficient on D_North means "compared to the reference group". Always state the reference category explicitly in your interpretation.
14. Using levels when both series are I(1)	If both Y and X are non-stationary I(1) series, use first differences (unless cointegration is established — not typically covered at introductory level).
15. Wrong n in LM formula	The auxiliary regression loses one observation for the lagged residual. Use $n_{\text{aux}} = n - 1$, or read from the ANOVA table: $n_{\text{aux}} = \text{Total df} + 1$.

3.2 Trick Phrasings — Same Question, Different Words

What the question writes	What it actually means
"Is X relevant to explaining Y?"	"Is β_X significantly different from zero?" → p-value test
"Does adding Region improve the model?"	"Are the region dummies jointly significant?" → joint F-test
"Baseline group" / "comparison group"	"Reference category" — the omitted dummy group
"The disturbance term carries memory"	"There is autocorrelation" → LMSC test
"Series wanders without returning to its mean"	"Series has a unit root" → DF test, then first differences
"Spurious correlation risk"	"Series is non-stationary" → run DF before modelling
"Long-run equilibrium effect"	"Total multiplier" → $\sum \beta_k$ or $\beta/(1-\gamma)$
"Impact effect of a price shock"	"Immediate effect" → β_0 only (coefficient on current X)
"Error variance is not constant"	"Heteroskedasticity" → White test
"Predictors are too correlated"	"Multicollinearity" → VIF and Tolerance
"Coefficient on lagged Y is 0.7"	" $\gamma = 0.7$ in AR(1)" → total multiplier = $\beta/(1-0.7) = \beta/0.3$
"Series is integrated of order one"	"I(1) — unit root in levels, stationary in first differences"
"Misspecification bias"	"Omitted variable bias" → $\text{sign}(\beta_Z) \times \text{sign}(r(X, Z))$
"First-order autocorrelation"	" $\varepsilon_t = \rho\varepsilon_{t-1} + u_t$ " → run LMSC test

QUICK DECISION GUIDE

MLR — Method Selector

Question type	Method	Key formula / output
Interpret coeff. of $\ln(X)$	Linear-log rule	$\beta/100$ units per 1% rise in X
Test one coefficient	Individual t-test	p-value (Sig.) in Coefficients table
Test a group of variables	Joint F-test	$F = [(SSE_r - SSE_u)/J] / [SSE_u/(n-k-1)]$
Check equal error variance	White test	$W = n \times R^2_{aux} \sim \chi^2(df)$
Direction of missing-variable distortion	Omitted variable bias	$\text{sign}(\beta_Z) \times \text{sign}(r(X,Z))$
Predictor correlation severity	VIF / Tolerance	$VIF = 1/(1-R^2_{aux})$, threshold = 5
Write equation with categories	Dummy variables	$k-1$ dummies, state reference group

Time Series — Method Selector

Question type	Method	Key output / decision
Is series stationary?	Dickey-Fuller test	$t < -2.86$ (V1) or -3.41 (V2) → stationary
Series drifts with time	DF test → unit root?	Unit root confirmed → difference it. No unit root → add t
Choose number of lags	Top-down DL selection	Drop highest lag while $p > 0.05$
Long-run effect of X on Y	Total multiplier	DL: $\sum \beta_k$ AR: $\beta/(1-\gamma)$
Immediate effect of X on Y	Impact multiplier	β_0 — coefficient on current X_t only
Check for autocorrelation	LMSC test	$LM = n \times R^2_{aux}$. Compare to 3.841
Model contains Y_{t-1}	Use LMSC, not DW	DW is invalid for AR models

✓ Exam Tips

- Always state H_0 and H_1 before every test — even when they seem obvious.
- Always write a full conclusion sentence: 'Since $p = X < 0.05$, we reject H_0 and conclude...'
- For DF: compare t-statistic (NOT p-value) to special critical value -2.86 or -3.41 .
- For LMSC and White test: use p-value directly (standard χ^2 distribution applies).
- For individual t-tests and F-tests in regression: use p-value from SPSS output.
- When unsure about stationarity: use DF Version 2 (with trend) — it is the safer default.
- Total multiplier always refers to the LONG-RUN effect after a PERMANENT change in X .